Appello 4 -Soluzione

venerdì 11 settembre 2015 10:40

Analisi

1) $f(x,y) = \frac{x^2 y}{2} - x^2 + xy - 2x - y^2$

a) $\nabla f(x,y) = \left(xy - 2x + y - 2,\ \frac{x^2}{2} + x - 2y\right)$

$\nabla f(x,y) = 0 \iff \begin{cases} xy - 2x + y - 2 = 0 & (1) \\ x^2 + 2x - 4y = 0 & (2) \end{cases}$

(1): $y(x+1) = 2(x+1)$: se $x = -1$ le (2): $y = -1/4$

Se $x \neq -1$: $y = 2$ (2) $\Rightarrow$ $x^2 + 2x - 8 = 0$

$\Delta' = 9$, $x = -1 \pm 3 \begin{cases} -4 \\ 2 \end{cases}$

Pti critici: $(-4,2)$, $(2,2)$, $(-1,-1/4)$.

$$\text{Hess} f(x,y) = \begin{pmatrix} y-2 & x+1 \\ x+1 & -2 \end{pmatrix}$$

$\text{Hess} f(2,2) = \begin{pmatrix} 0 & 3 \\ 3 & -2 \end{pmatrix}$ det $\text{Hess} f(2,2) = -9 < 0$: sella

$\text{Hess} f(-4,2) = \begin{pmatrix} -6 & 3 \\ 3 & -2 \end{pmatrix}$ det $\text{Hess} f(-4,2) = 12 - 9 < 0$: sella

$\text{Hess} f(-1,-\frac{1}{4}) = \begin{pmatrix} -11/4 & 3/4 \\ 3/4 & -2 \end{pmatrix}$ $\begin{cases} \det \text{Hess} f(-1,-\frac{3}{4}) = \frac{22}{4} - \frac{9}{16} > 0 \\ -\frac{11}{4} < 0: \quad (-1,-\frac{1}{4}) \text{ MAX locale stretto.} \end{cases}$

b) $\nabla f(0,1) = (-1,-2)$

$z = \nabla f(0,1) \cdot \begin{pmatrix} x-0 \\ y-1 \end{pmatrix} + f(0,1)$

$$= \begin{pmatrix} -1 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} x \\ y-1 \end{pmatrix} - 1 = -x - 2(y-1) - 1$$

$$\boxed{z = -x - 2y + 1}$$

c) $$\partial_{\vec{u}} f(0,1) = \nabla f(0,1) \cdot \vec{u} = \begin{pmatrix} -1 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{pmatrix} = -\frac{\sqrt{3}}{2} - 1 .$$

2. $$\begin{cases} y' + (1-x)y = xe^{-x} \\ y(1) = 0 \end{cases}$$

$$\int 1-x \, dx = x - \frac{x^2}{2} \qquad \left[y' + (1-x)y\right] e^{x - \frac{x^2}{2}} = x e^{-\frac{x^2}{2}}$$

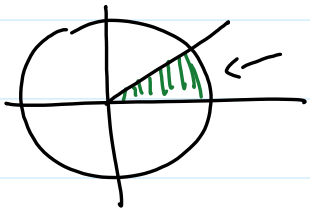
$$\left(y e^{x - \frac{x^2}{2}}\right)' = x e^{-\frac{x^2}{2}}$$

$$\Rightarrow \quad y e^{x - \frac{x^2}{2}} = \int x e^{-\frac{x^2}{2}} dx = -e^{-\frac{x^2}{2}} + c$$

$$y(x) = -e^{-x} + c e^{-x + \frac{x^2}{2}} . \quad \text{e} \quad y(1) = 0 \iff -e^{-1} + c e^{-\frac{1}{2}} = 0$$

$$\iff c = e^{-\frac{1}{2}}$$

$$\Rightarrow \boxed{y(x) = -e^{-x} + e^{-x + \frac{x^2}{2} - \frac{1}{2}}}$$

3. 

$(\rho \cos t, \rho \sin t)$: $\rho \in [0,1]$, $t \in [0, \pi/4]$.

$$\int_D \frac{x+y}{1+x^2+y^2} dx\, dy = \int_{\substack{0 \le \rho \le 1 \\ t \in [0, \pi/4]}} \frac{\rho \cos t + \rho \sin t}{1 + \rho^2} \rho \, d\rho \, dt$$

$$= \left(\int_0^1 \frac{\rho^2}{1+\rho^2} d\rho \right) \left(\int_0^{\pi/4} \cos t + \sin t \, dt \right)$$

$$\int_0^1 \frac{\rho^2}{1+\rho^2} d\rho = \int_0^1 1 - \frac{1}{1+\rho^2} d\rho = 1 - \left[\operatorname{arctg} \rho\right]_0^1 = 1 - \frac{\pi}{4}$$

$$\int_0 \frac{1}{1+\rho^2}\,d\rho = \int_0 1 - \frac{\quad}{1+\rho^2}\,d\rho = 1 - [\text{arctg}(\rho)]_0^{\quad} = \frac{\pi}{4}$$

$$\int_0^{\pi/4} \cos t + \sin t \, dt = \left[\sin t - \cos t\right]_0^{\pi/4} = 1$$

$\Rightarrow$ l'integrale vale $\boxed{1 - \frac{\pi}{4}}$

Probabilità

1) a) A: allarme B: contiene una bomba

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$
$$= 0.99 \cdot \frac{1}{5000} + 0.05 \cdot \frac{4999}{5000}$$

b) $$P(B|A) = \frac{P(A|B)P(B)}{P(A)} = \frac{0.99}{0.99 + 0.05 \cdot 4999}$$

2) a) $X \sim B\left(1000, \frac{18}{100}\right)$

b) $P(X \le 170) + P(X \ge 190)$

$$X \approx 1000 \times \frac{18}{100} + \sqrt{1000 \times \frac{18}{100} \times \frac{82}{100}}\; Z$$
$$= 180 + \underbrace{\sqrt{\frac{18 \times 82}{10}}}_{\sigma}\; Z \qquad Z \sim N(0,1)$$

$$P(X \le 170) \approx P(180 + \sigma Z \le 170) = P\left(Z \le \frac{-10}{\sigma}\right)$$
$$= \Phi\left(-\frac{10}{\sigma}\right) = 1 - \Phi\left(\frac{10}{\sigma}\right)$$

$$P(X \ge 190) \approx P(180 + \sigma Z \ge 190)$$
$$= P(\sigma Z \ge 10) = P\left(Z \ge \frac{10}{\sigma}\right) = 1 - \Phi\left(\frac{10}{\sigma}\right)$$

Quindi $P(X \le 170) + P(X \ge 190) \approx \boxed{2\left(1 - \Phi\left(\frac{10}{\sigma}\right)\right)}$ $\quad \sigma = \sqrt{\frac{18 \times 82}{10}} = 6 \times \sqrt{\frac{41}{10}}$

$+\infty \qquad , 0 \text{ se } x < 0$

$-6\sqrt{10}$

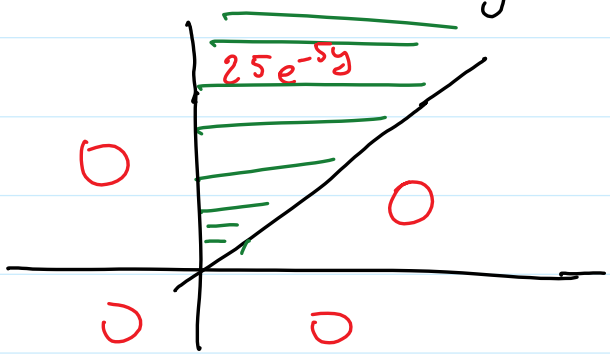
3. a) $$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y)\,dy = \begin{cases} 0 & \text{se } x<0 \\ \displaystyle\int_x^{+\infty} 25\, e^{-5y}\,dy & \text{se } x\geq 0 \end{cases}$$

$$= -\frac{25}{5}\left[e^{-5y}\right]_x^{+\infty} = 5e^{-5x}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y)\,dx = \begin{cases} 0 & \text{se } y<0 \\ \displaystyle\int_0^{y} 25\, e^{-5y}\,dx & \text{se } y\geq 0 \end{cases}$$

$$= 25y\,e^{-5y}$$

(b) $f_{X,Y}(x,y)$:

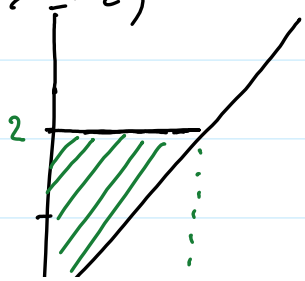


X, Y non indipendenti perché $f_X(x)\,f_Y(y) \neq f_{X,Y}(x,y)$.

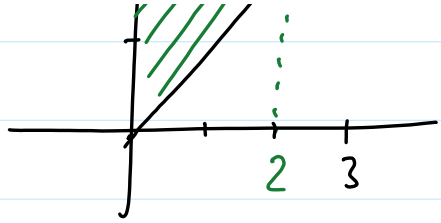
Ad esempio se $x>0, y>0$ è $f_X(x)\,f_Y(y) \neq 0$ mentre $f_{X,Y}(x,y)=0$ se $y>x>0$.

(c) $P(X\leq 3, Y\leq 2) = P(0\leq X\leq 3,\ 0\leq Y\leq 2)$

$$= \int 25\, e^{-5y}\,dx\,dy =$$



$$[0,3]\times[0,2]\cap\{0\le x<y\}$$



$$= \int_{[0,2]\times[0,2]\cap\{0\le x<y\}} 25\, e^{-5y}\, dx\, dy = \int_0^2 \left\{ \int_0^y 25 e^{-5y}\, dx \right\} dy$$

$$= \int_0^2 25\, y e^{-5y}\, dy \underset{\text{p. parti}}{=} 25\left(\left[-\frac{1}{5} y e^{-5y} \right]_0^2 + \frac{1}{5}\int_0^2 e^{-5y}\, dy \right)$$

$$= 25\left(-\frac{2}{5} e^{-10} - \frac{1}{25}\left(e^{-10}-1\right)\right)$$

$$= -10e^{-10} - e^{-10} + 1 = \boxed{1 - 11 e^{-10}}$$